

External Controls — Column Shift

 [Ford Ecat Electronic Parts Catalog](#)

Special Tool(s)

 ST3093-A	Fluke 77-IV Digital Multimeter FLU77-4 or equivalent
 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool

Inspection and Verification

1. Verify the customer concern by operating the transmission external control.
2. Visually inspect for obvious signs of mechanical and electrical damage; refer to the following chart:

Visual Inspection Chart

Mechanical	Electrical
• Damaged Brake Shift Interlock Actuator (BSIA) • Damaged Tow/Haul switch • Damaged SelectShift™ switch	• Failed fuse(s): ▪ Body Control Module (BCM) 18 (10A) ▪ BCM 42 (5A) • PCM • Loose or damaged connections • Damaged wiring harness • Corroded or damaged terminals

3. If the fault is not visually evident, determine the symptom. GO to [Symptom Chart — External Controls, Column Shift](#) or GO to [Symptom Chart — NVH Column Shift](#).

DTC Chart

DTC Chart

DTC	Component	Description	Condition	Symptom	Action
B1319:11	Brake Shift Interlock Actuator (BSIA) , BCM	Shift Interlock (BSI) Output Circuit: Circuit short to ground	BSIA system malfunction	BSIA inoperative.	GO to Pinpoint Test A.
B1319:15	Brake Shift Interlock Actuator (BSIA) , BCM	Shift Interlock (BSI) Output Circuit: Circuit short to voltage or open	BSIA system malfunction	BSIA inoperative.	GO to Pinpoint Test A.
P0815	Upshift Switch Circuit	Upshift switch concerns, open or shorted circuit	The PCM signal input voltage status is out of range	Upshift switch inoperative.	GO to Pinpoint Test D.
P0816	Downshift Switch Circuit	Downshift switch concerns, open or shorted circuit	The PCM signal input voltage status is out of range	Downshift switch inoperative.	GO to Pinpoint Test D.
P1780	Tow/Haul Switch	Transmission Control Switch (TCS) (Overdrive (O/D) cancel) circuit out of self-test range	The PCM did not receive a Tow/Haul switch input during the Key ON Engine Running (KOER) self-test	Tow/Haul is inoperative or does not operate correctly.	GO to Pinpoint Test C.

Symptom Chart — External Controls, Column Shift

Symptom Chart — External Controls, Column Shift

Condition	Possible Causes	Action
The Brake Shift Interlock Actuator (BSIA) system does not release/lock correctly	- Circuitry - Fuse(s): - Body Control Module (BCM) 18 (10A) - BSIA - Selector lever cable	GO to Pinpoint Test A. - VERIFY selector lever cable adjustment. REFER to Selector Lever Cable Adjustment — Column Shift in this section.
Transmission Range (TR) indicator (PRNDL) does not correspond to the selected gear	- Selector lever cable bracket - Instrument Panel Cluster (IPC) malfunction - Selector lever cable loose in the selector lever cable bracket	- TIGHTEN bolts holding the selector lever cable bracket. REFER to Selector Lever Cable Adjustment — Column Shift in this section. - REFER to Section 413-01. - VERIFY cable is fully seated in the selector lever cable bracket. REFER

	• Selector lever cable • Clip securing the selector lever cable to the steering column bracket	to Selector Lever Cable — Column Shift in this section. • VERIFY selector lever cable adjustment. REFER to Selector Lever Cable Adjustment — Column Shift in this section. • INSTALL clip correctly. REFER to Selector Lever Cable — Column Shift in this section.
• TR indicator (PRNDL) does not illuminate	• IPC malfunction • TR sensor	• REFER to Section 413-00. • CHECK for TR sensor DTCs. REFER to Section 307-01.
• The Tow/Haul switch does not operate correctly	• Tow/Haul switch • Tow/Haul switch not cycled during self-test • Circuitry	• GO to Pinpoint Test C
• The Tow/Haul indicator lamp is not operating correctly	• IPC • PCM • Circuitry	• REFER to Section 413-01.
• The SelectShift™ switch is not operating correctly	• SelectShift™ switch • PCM • Circuitry	• GO to Pinpoint Test D.
• Water enters inside the vehicle	• Cable assembly grommet • Torn selector lever cable assembly grommet	• SECURE grommet to floor panel. REFER to Selector Lever Cable — Column Shift in this section. • INSTALL a new selector lever cable. REFER to Selector Lever Cable — Column Shift in this section.
• Excessive shift effort	• Selector lever cable • Cable bracket	• INSTALL a new selector lever cable. REFER to Selector Lever Cable — Column Shift in this section. • ADJUST cable. REFER to Selector Lever Cable Adjustment — Column Shift in this section. TIGHTEN cable bracket screws. REFER to Specifications in this section.

• Selector lever will not shift from PARK	• Selector lever cable • Column bracket clip • Broken selector lever cable • Brake Shift Interlock Actuator (BSIA) system failure	• VERIFY selector lever cable adjustment. REFER to Selector Lever Cable Adjustment — Column Shift in this section. • INSTALL a new bracket clip. REFER to Selector Lever Cable — Column Shift in this section. • INSTALL a new selector lever cable. REFER to Selector Lever Cable — Column Shift in this section. • GO to Pinpoint Test A.
• Vehicle will not crank/start in park but will crank/start in neutral	• Selector lever cable • TR sensor	• VERIFY selector lever cable adjustment. REFER to Selector Lever Cable Adjustment — Column Shift in this section. • CHECK for TR sensor DTCs. REFER to Section 307-01.

Symptom Chart — NVH Column Shift

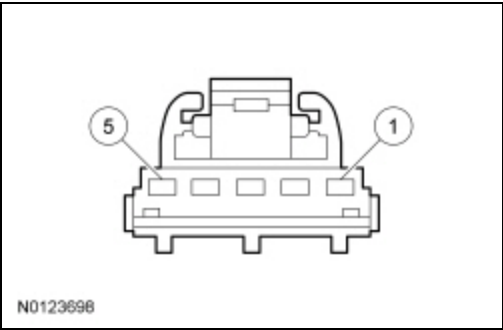
Symptom Chart — NVH Column Shift		
Condition	Possible Causes	Action
NOTE: NVH symptoms should be identified using the diagnostic tools that are available. For a list of these tools, an explanation of their uses and a glossary of common terms, refer to Section 100-04. Since it is possible any one of multiple systems may be the cause of a symptom, it may be necessary to use a process of elimination type of diagnostic approach to pinpoint the responsible system. If this is not the causal system for the symptom, refer back to Section 100-04 for the next likely system and continue diagnosis.		
• Vibration — a high frequency (20-80 Hz) that is felt through the seat or selector lever. Changes with engine speed	• Selector lever cable incorrectly routed, grounded out or loose	• CHECK the selector lever cable. REPAIR as necessary. REFER to Selector Lever Cable — Column Shift in this section.
• Rattle, buzz or other noise	• Selector lever loose • Selector lever • Cable assembly grommet	• TIGHTEN the selector lever bolts. REFER to Selector Lever — Column Shift in this section. • INSTALL a new selector lever. REFER to Selector Lever — Column Shift in this section. • SECURE the grommet to the floor panel. REFER to Selector Lever Cable — Column Shift in this section.

Pinpoint Tests

Refer to Wiring Diagrams Cell [30](#) , Transmission Controls for schematic and connector information.

Refer to Wiring Diagrams Cell [37](#) , Shift Interlock for schematic and connector information.

Tow/Haul Connector — C2399



Pin Number	Circuit Function
1	Tow/Haul Power
2	Tow/Haul Signal
3	SelectShift™ Upshift
4	SelectShift™ Ground
5	SelectShift™ Downshift

PINPOINT TEST A : THE BSIA SYSTEM DOES NOT RELEASE/LOCK CORRECTLY

NOTE: *Inspect the selector lever cable before proceeding with this pinpoint test.*

A1 CHECK THE STOPLAMPS

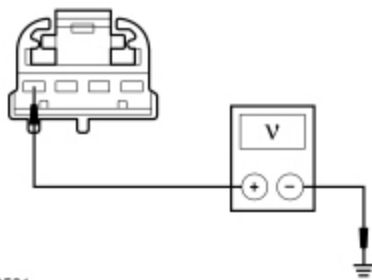
- Apply the brake pedal and observe the stoplamps.

Do the stoplamps illuminate?

Yes	VERIFY the BCM fuse 18 (10A) is OK, If OK GO to A2 . If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short.
No	REFER to Section 417-01.

A2 CHECK THE BSIA OUTPUT CIRCUIT

- Disconnect: Brake Shift Interlock Actuator (BSIA) [C2008](#) .
- Ignition ON.
- Measure the voltage between the BSIA [C2008](#) Pin 4, circuit CET53 (BU/OG), harness side and ground.



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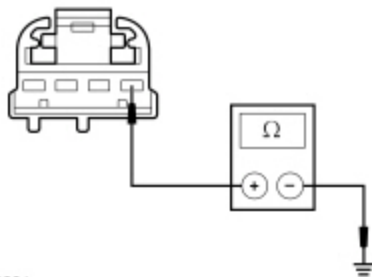
- Depress the brake pedal.
- With the brake pedal depressed, measure the voltage between the BSIA [C2008](#) Pin 4, circuit CET53 (BU/OG), harness side and ground.

Is the voltage greater than 10 volts with the brake pedal applied and 0 volts with the brake pedal not applied?

Yes	GO to A3 .
No	If the voltage was greater than 10 volts with the brake pedal applied or not applied, GO to A4 . Otherwise, GO to A5 .

A3 CHECK THE BSIA GROUND CIRCUIT FOR AN OPEN

- Ignition OFF.
- Measure the resistance between the BSIA [C2008](#) Pin 1, circuit GD133 (BK), harness side and ground.



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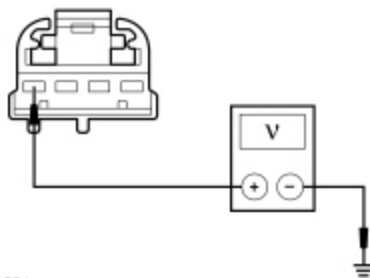
Is the resistance less than 5 ohms?

Yes	REPLACE the BSIA . REFER to Brake Shift Interlock Actuator — Column Shift in this section. CLEAR the DTC. TEST the system for normal operation.
No	REPAIR circuit GD133 (BK) for an open. CLEAR the DTC. TEST the system for normal operation.

A4 CHECK THE BSIA OUTPUT CIRCUIT FOR A SHORT TO POWER

- Ignition OFF.
- Disconnect: BCM [C2280B](#) .
- Inspect [C2008](#) and [C2280B](#) for damaged, pushed-out terminals or loose wires.
- Ignition ON.

- Measure the voltage between the BSIA [C2008](#) Pin 4, circuit CET53 (BU/OG), harness side and ground.



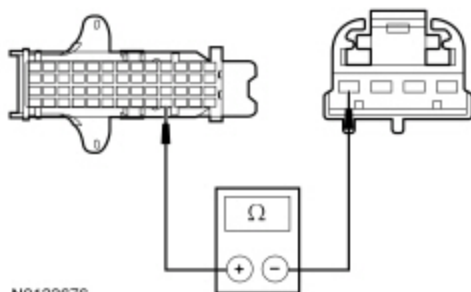
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Is any voltage present?

Yes	REPAIR circuit CET53 (BU/OG) for a short to power. CLEAR the DTC. TEST the system for normal operation.
No	GO to A5 .

A5 CHECK THE BSIA POWER CIRCUIT FOR AN OPEN

- Measure the resistance between the BSIA [C2008](#) Pin 4, circuit CET53 (BU/OG) and BCM [C2280B](#) Pin 49, CET53 (BU/OG) harness side.



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Is the resistance less than 5 ohms?

Yes	GO to A6 .
No	REPAIR circuit CET53 (BU/OG) for an open. CLEAR the DTC. TEST the system for normal operation.

A6 CHECK FOR CORRECT BCM OPERATION

- Disconnect all BCM connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the BCM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	REPLACE the BCM . REFER to Section 418-00. CLEAR the DTC. TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

PINPOINT TEST C : THE TOW/HAUL SWITCH DOES NOT OPERATE CORRECTLY

NOTE: Vehicles equipped with a 3.7L engine use PCM [C175T](#).

NOTE: Vehicles equipped with a 5.0L or 6.2L engine use PCM [C138IT](#).

NOTE: Vehicles equipped with a 3.5L engine use PCM [C1551B](#).

C1 CHECK DTCS

- Ignition ON.
- Connect the scan tool.
- Start the engine and RUN the Key ON Engine Running (KOER) self-test.

Is DTC P1780 set?

Yes	GO to C2 .
No	No trouble found with the Tow/Haul system. REFER to Section 413-01 if the Tow/Haul lamp does not illuminate.

C2 CHECK THE BCM FUSE 42 (5A)

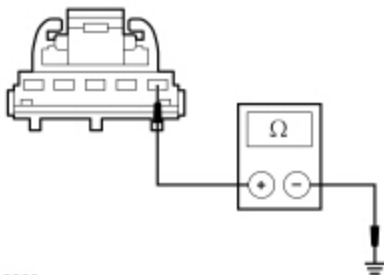
- Ignition OFF.
- Remove Body Control Module (BCM) fuse 42 (5A).
- Check fuse: BCM 42 (5A) .

Is the resistance less than 5 ohms?

Yes	GO to C4 .
No	GO to C3 .

C3 CHECK THE TOW/HAUL POWER CIRCUIT FOR A SHORT TO GROUND

- Disconnect: Tow/Haul [C2399](#).
- Disconnect: BCM [C2280B](#) .
- Inspect [C2399](#) and [C2280B](#) for damaged, pushed-out terminals or loose wires.
- Measure the resistance between the Tow/Haul [C2399](#) Pin 1, circuit CBP42 (GN), harness side and ground.



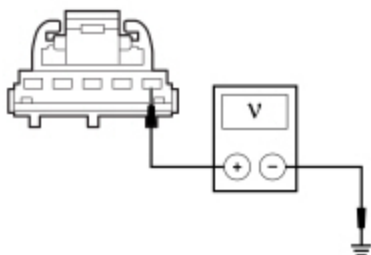
N0112233

Is the resistance greater than 10,000 ohms?

Yes	GO to C4 .
No	REPAIR circuit CBP42 (GN) for a short to ground. CLEAR the DTC. RERUN the KOER self-test and test the system for normal operation.

C4 CHECK THE TOW/HAUL POWER CIRCUIT FOR VOLTAGE

- Disconnect: Tow/Haul [C2399](#).
- Inspect [C2399](#) for damaged, pushed-out terminals or loose wires.
- Ignition ON.
- Measure the voltage between the Tow/Haul switch [C2399](#) Pin 1, circuit CBP42 (GN), harness side and ground.



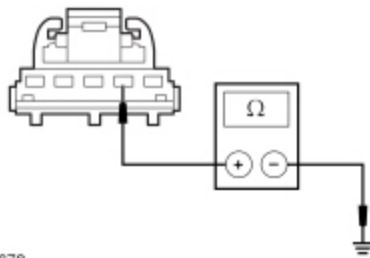
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Is the voltage greater than 10 volts?

Yes	GO to C5 .
No	REPAIR circuit CBP42 (GN) for an open. CLEAR the DTC. RERUN the KOER self-test and test the system for normal operation.

C5 CHECK THE TOW/HAUL INPUT CIRCUIT FOR A SHORT TO GROUND

- Disconnect: PCM [C175T](#).
- Disconnect: PCM [C1381T](#).
- Disconnect: PCM [C1551B](#).
- Inspect the connector for damaged, pushed-out terminals, corrosion or loose wires.
- Measure the resistance between Tow/Haul switch [C2399](#) Pin 2, circuit CET34 (BN/GN) and ground.



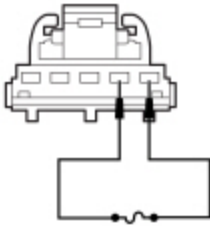
N0123678

Is the resistance greater than 10,000 ohms?

Yes	GO to C6 .
No	REPAIR circuit CET34 (BN/GN) for a short to ground. REPLACE the BCM 42 (5A). CLEAR the DTC. RERUN the KOER self-test and test the system for normal operation.

C6 CHECK THE TOW/HAUL INPUT CIRCUIT FOR AN OPEN

- Connect the scan tool.
- Enter the following diagnostic mode on the scan tool: TCS_DEPRES .
- Connect a fused jumper from Tow/Haul switch, [C2399](#) Pin 1 to [C2399](#) Pin 2, harness side.



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Does the TCS_DEPRES PID display YES?

Yes	REPLACE the selector lever. REFER to Selector Lever — Column Shift in this section. CLEAR the DTC. RERUN the KOER self-test and test the system for normal operation.
No	INSPECT and REPAIR circuit CET34 (BN/GN) for an open. If an open circuit is not found, REPLACE the PCM. PERFORM the Solenoid Body Strategy Data Download procedure. REFER to Section 307-01. RERUN the KOER self-test and test the system for normal operation.

PINPOINT TEST D : THE SELECTSHIFT™ SWITCH IS NOT OPERATING CORRECTLY

NOTE: Vehicles equipped with a 3.7L engine use PCM [C175B](#).

NOTE: Vehicles equipped with a 5.0L or 6.2L engine use PCM [C1381B](#).

NOTE: Vehicles equipped with a 3.5L engine use PCM [C1551B](#).

D1 CHECK DTCS

- Ignition ON.

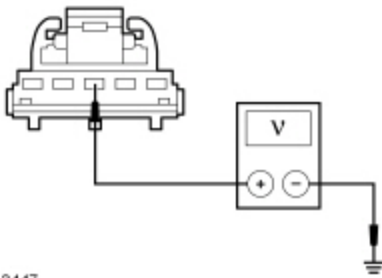
- Connect the scan tool.
- Start the engine and RUN the Key ON Engine Running (KOER) self-test.

Is DTC P0815 set?

Yes	GO to D2 .
No	GO to D5 .

D2 CHECK THE UPSHIFT CIRCUIT FOR VOLTAGE

- Ignition OFF.
- Disconnect: Tow/Haul [C2399](#).
- Inspect [C2399](#) for damaged, pushed-out terminals, corrosion or loose wires.
- Ignition ON.
- Measure the voltage between the Tow/Haul switch [C2399](#) Pin 3, circuit CET43 (GY), harness side and ground.



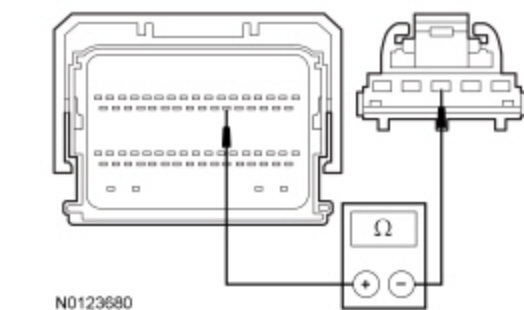
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Is the voltage greater than 10 volts?

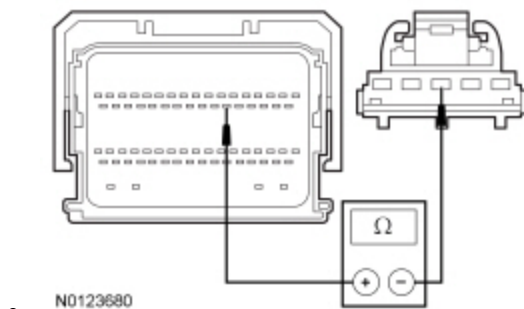
Yes	GO to D4 .
No	GO to D3 .

D3 CHECK THE UPSHIFT CIRCUIT FOR AN OPEN

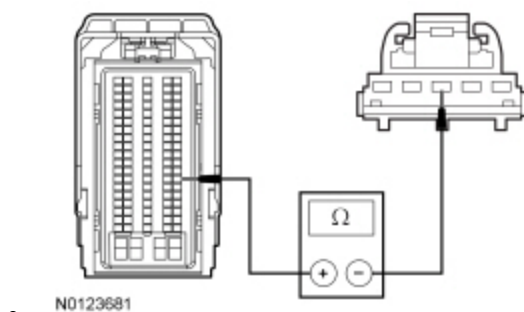
- Ignition OFF.
- Disconnect: PCM [C175B](#).
- Disconnect: PCM [C1381B](#).
- Disconnect: PCM [C1551B](#).
- Inspect the connector for damaged, pushed-out terminals, corrosion or loose wires.
- For 3.7L engine vehicles, measure the resistance between the Tow/Haul switch [C2399](#) Pin 3, circuit CET43 (GY) and PCM [C175B](#) Pin 23, harness side.



- For 5.0L or 6.2L engine vehicles, measure the resistance between the Tow/Haul switch [C2399](#) Pin 3, circuit CET43 (GY) and PCM [C1381B](#) Pin 23, harness side.



- For 3.5L engine vehicles, measure the resistance between the Tow/Haul switch [C2399](#) Pin 3, circuit CET43 (GY) and PCM [C1551B](#) Pin 89, harness side.

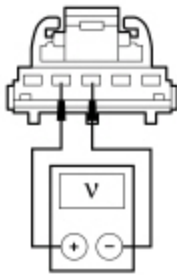


Is the resistance less than 5 ohms?

Yes	GO to D4 .
No	REPAIR circuit CET43 (GY) for an open. CLEAR the DTCs. RERUN the KOER self-test and test the system for normal operation.

D4 CHECK THE SIGNAL RETURN CIRCUIT FOR AN OPEN

- Connect: PCM [C175B](#).
- Connect: PCM [C1381B](#).
- Connect: PCM [C1551B](#).
- Ignition ON.
- Measure the voltage between Tow/Haul switch [C2399](#) Pin 3 and [C2399](#) Pin 4 harness side.



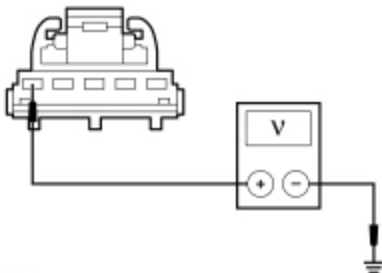
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Is the voltage greater than 10 volts?

Yes	INSTALL a new selector lever. REFER to Selector Lever — Column Shift in this section. CLEAR the DTCs. RERUN the KOER self-test and test the system for normal operation.
No	INSPECT and REPAIR the signal return circuit RE407 (YE/VT) for an open. If an open circuit is not found, REPLACE the PCM. PROGRAM the PCM with the latest calibration level. PERFORM the solenoid strategy download procedure, REFER to solenoid body strategy in Section 307-01. RERUN the KOER self-test and test the system for normal operation.

D5 CHECK THE DOWNSHIFT CIRCUIT FOR VOLTAGE

- Ignition OFF.
- Disconnect: Tow/Haul [C2399](#).
- Inspect [C2399](#) for damaged, pushed-out terminals, corrosion or loose wires.
- Ignition ON.
- Measure the voltage between the Tow/Haul switch [C2399](#) Pin 5, circuit CET42 (GN/VT), harness side and ground.



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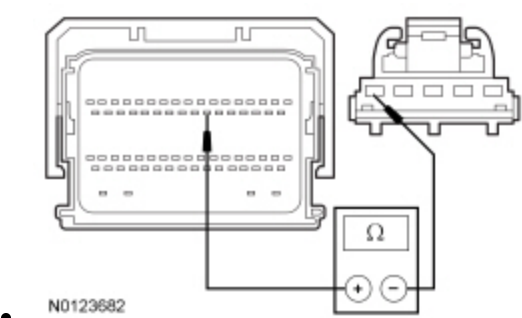
Is the voltage greater than 10 volts?

Yes	GO to D7 .
No	GO to D6 .

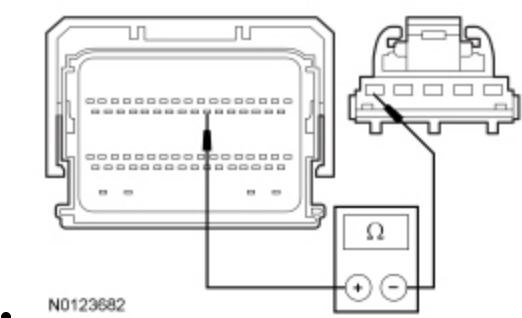
D6 CHECK THE DOWNSHIFT CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: PCM [C175B](#).

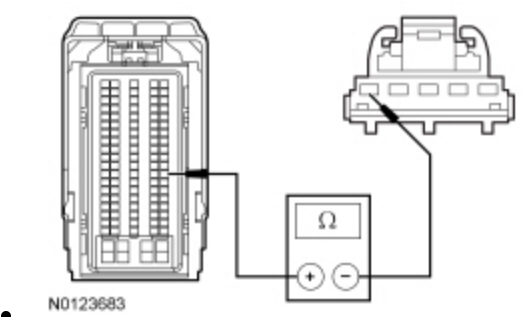
- Disconnect: PCM [C1381B](#).
- Disconnect: PCM [C1551B](#).
- Inspect the connector for damaged, pushed-out terminals, corrosion or loose wires.
- For 3.7L engine vehicles, measure the resistance between the Tow/Haul switch [C2399](#) Pin 5, circuit CET42 (GN/VT) and PCM [C175B](#) Pin 24, harness side.



- For 5.0L or 6.2L engine vehicles, measure the resistance between the Tow/Haul switch [C2399](#) Pin 5, circuit CET42 (GN/VT) and PCM [C1381B](#) Pin 24, harness side.



- For 3.5L engine vehicles, measure the resistance between the Tow/Haul switch [C2399](#) Pin 5, circuit CET42 (GN/VT) and PCM [C1551B](#) Pin 88, harness side.



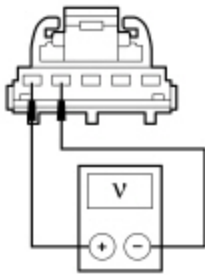
Is the resistance less than 5 ohms?

Yes	GO to D7 .
No	REPAIR circuit CET42 (GN/VT) for an open. CLEAR the DTCs. RERUN the KOER self-test and test the system for normal operation.

D7 CHECK THE SIGNAL RETURN CIRCUIT FOR AN OPEN

- Connect: PCM [C175B](#).
- Connect: PCM [C1381B](#).

- Connect: PCM [C1551B](#).
- Ignition ON.
- Measure the voltage between tow/haul switch [C2399](#) Pin 5 and [C2399](#) Pin 4 harness side.



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Is the voltage greater than 10 volts?

Yes	INSTALL a new selector lever. REFER to Selector Lever — Column Shift in this section. CLEAR the DTCs. RERUN the KOER self-test and test the system for normal operation.
No	INSPECT and REPAIR the signal return circuit RE407 (YE/VT) for an open. If an open circuit is not found, REPLACE the PCM. PROGRAM the PCM with the latest calibration level. PERFORM the solenoid strategy download procedure, REFER to solenoid body strategy in Section 307-01. RERUN the KOER self-test and test the system for normal operation.